

Exp no :07

Date :

Use AFLogical OSE to Extract Data from an Android Device

Description

AFLogical OSE (Open Source Edition) is a tool designed for logical data extraction from Android devices. As part of the Open Source Android Forensics project, it collects non-file system data like contacts, messages (SMS/MMS), and call logs without needing root access.

Step 1: Prepare Your Environment

1. Download AFLogical OSE:

- Visit the AFLogical OSE GitHub page and download the latest release, or clone the repository using Git.

2. Install Java:

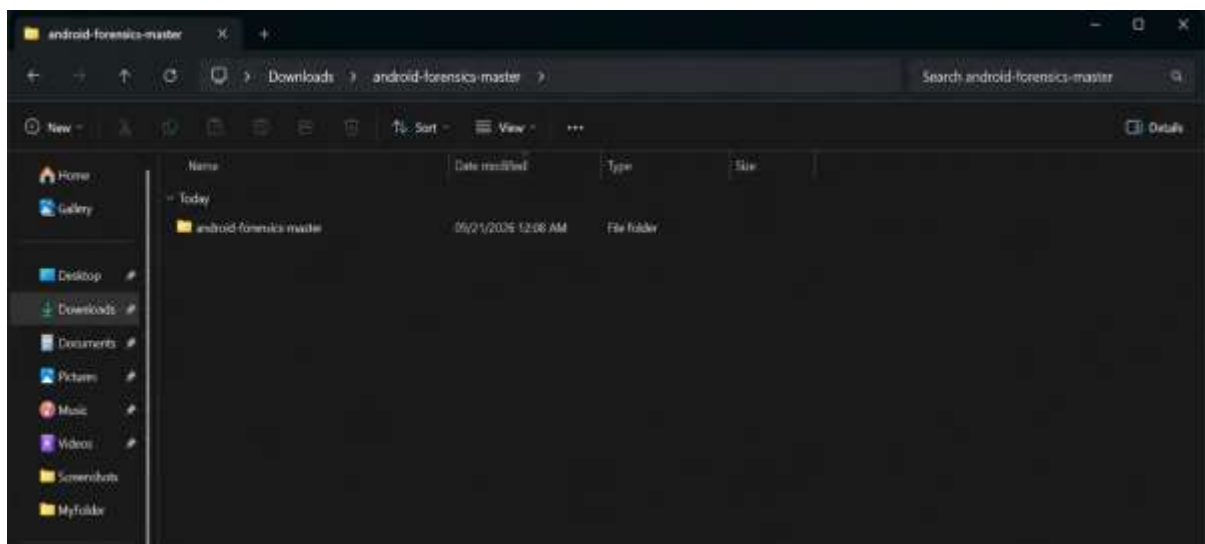
- AFLogical OSE requires Java to run. Ensure Java is installed on your machine.

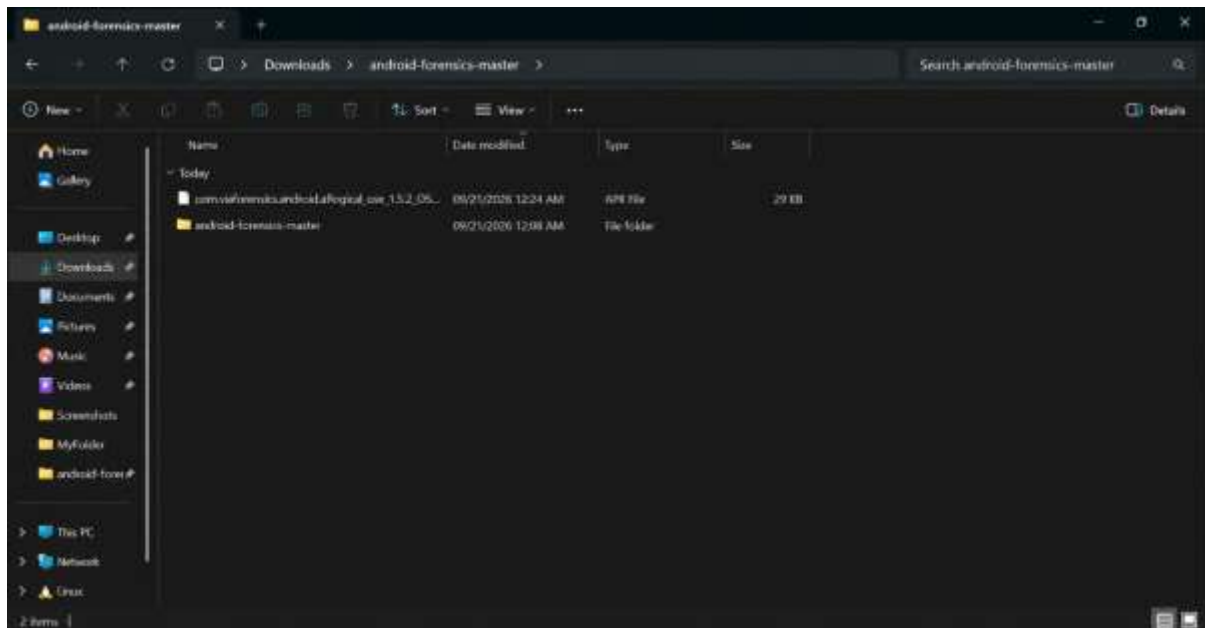
3. Install Android Debug Bridge (ADB):

- Download **ADB** (a command-line tool for device communication) from the Android Developer's website.
- Install it on your system and **add it to your system's PATH environment variable** for easy access from the command line.

4. Enable USB Debugging on the Android Device:

- Go to **Settings > About Phone** and tap **Build Number seven times** to enable Developer Options.
- Go to **Settings > Developer Options** and enable **USB Debugging**.





Step 2: Connect the Android Device to Your Computer

1. Connect via USB:

- Use a USB cable to connect the Android device to your computer.

2. Verify ADB Connection:

- Open Command Prompt/Terminal and run the following command:
`adb devices`
- **Verification:** The device should be listed, confirming that the ADB connection is ready.

```
C:\Users\duddu\Downloads\platform-tools-latest-windows\platform-tools>adb devices
List of devices attached
ZA22258HP5    unauthorized

C:\Users\duddu\Downloads\platform-tools-latest-windows\platform-tools>
```

Step 3: Extract Data Using AFLogical OSE

1. Push the APK to the Android Device:

- Navigate to the AFLogical OSE directory in your terminal.
- Run the command to install the AFLogical OSE APK on the device:

`adb install aflogical.apk`

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\duddu\Downloads\android-forensics-master> adb install con.viaforensics.android.aflogical_ose_1.5.2_OSE_152.apk
Performing Streamed Install
Success
PS C:\Users\duddu\Downloads\android-forensics-master>
```

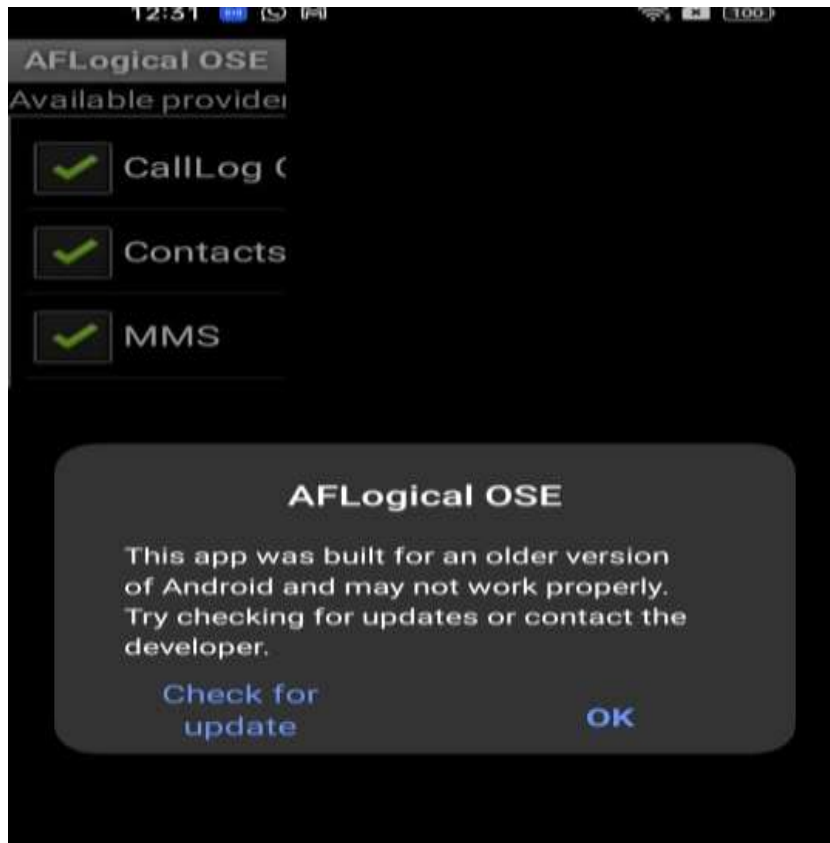
2. Launch the AFLogical OSE App on the Device:

- On the Android device, open the newly installed AFLogical app.

3. Select and Start Data Extraction:

- Select the desired data types (e.g., contacts, SMS, MMS, call logs) from the app's options.
- The app will start the extraction process, storing the data in **.csv files** on the device's storage, typically in a directory named aflogical.



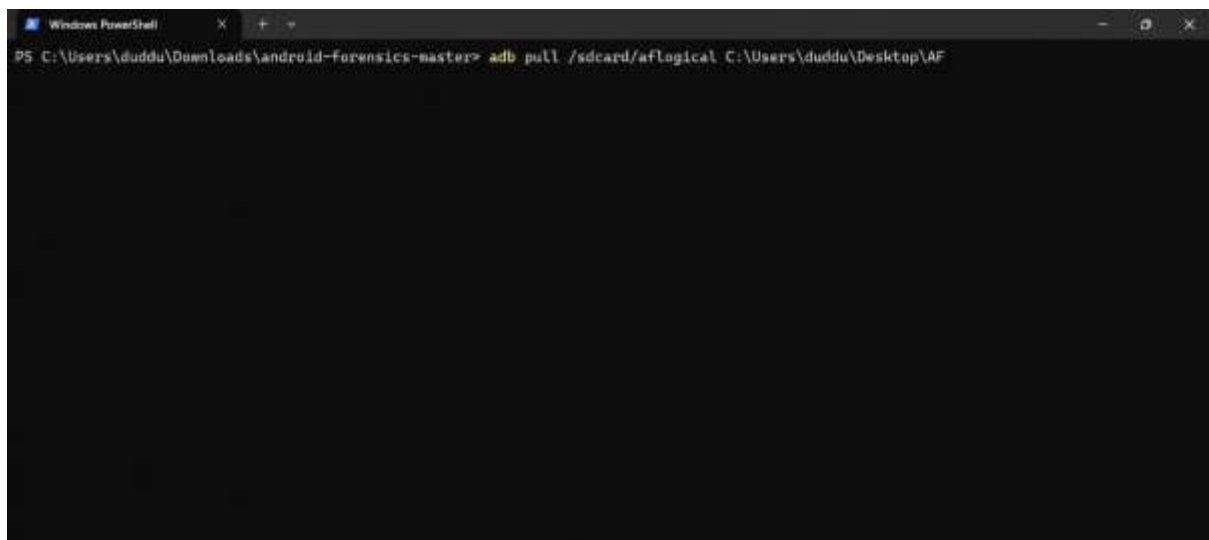


Step 4: Transfer Extracted Data to Your Computer

1. Use ADB to Pull Data:

- After extraction is complete, transfer the collected data files from the Android device to your computer using the adb pull command:

`adb pull /sdcard/aflogical /path/to/destination`



Note: Replace `/path/to/destination` with the desired save location on your computer.

2. Verify the Data:

- Navigate to the destination directory and check the .csv files to ensure all required data has been successfully transferred.

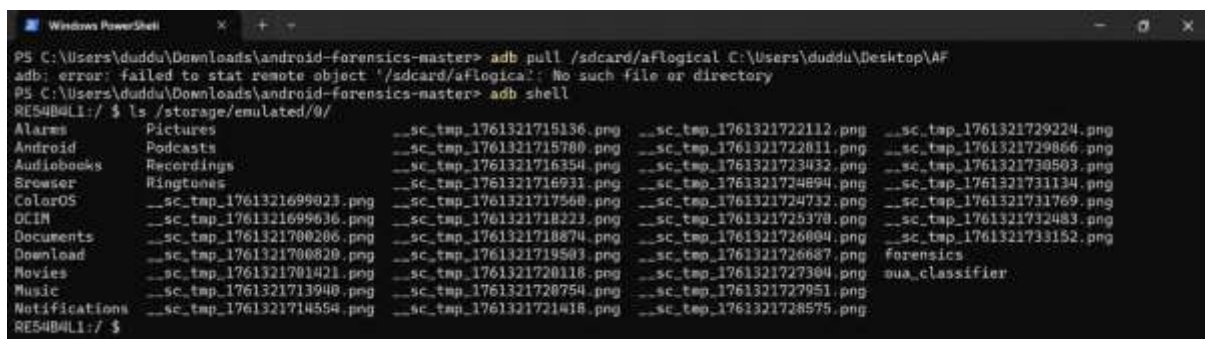
Step 5: Analyze the Extracted Data

1. Open the CSV Files:

- Use applications like **Excel**, **Google Sheets**, or a text editor to open and analyze the .csv files containing the extracted logical data.

2. Review and Document:

- Carefully review the data for any relevant evidence or information.
- Document all findings and prepare a comprehensive report.



```
PS C:\Users\duddu\Downloads\android-forensics-master> adb pull /sdcard/aflogical C:\Users\duddu\Desktop\AF
adb: error: failed to stat remote object '/sdcard/aflogical': No such file or directory
PS C:\Users\duddu\Downloads\android-forensics-master> adb shell
RES4B4LI:/ $ ls /storage/emulated/0/
Alarms      Pictures      _sc_tmp_1761321715136.png  _sc_tmp_1761321722112.png  _sc_tmp_1761321729224.png
Android     Podcasts      _sc_tmp_1761321715780.png  _sc_tmp_1761321722811.png  _sc_tmp_1761321729866.png
Audiobooks  Recordings   _sc_tmp_1761321716354.png  _sc_tmp_1761321723432.png  _sc_tmp_1761321730503.png
Browser     Ringtones    _sc_tmp_1761321716931.png  _sc_tmp_1761321724894.png  _sc_tmp_1761321731134.png
ColorOS     _sc_tmp_1761321699023.png  _sc_tmp_1761321717568.png  _sc_tmp_1761321724732.png  _sc_tmp_1761321731769.png
DCIM        _sc_tmp_1761321699636.png  _sc_tmp_1761321718223.png  _sc_tmp_1761321725378.png  _sc_tmp_1761321732483.png
Documents   _sc_tmp_1761321700206.png  _sc_tmp_1761321718874.png  _sc_tmp_1761321726004.png  _sc_tmp_1761321733152.png
Download    _sc_tmp_1761321700820.png  _sc_tmp_1761321719503.png  _sc_tmp_1761321726687.png  forensics
Movies      _sc_tmp_1761321701421.png  _sc_tmp_1761321720118.png  _sc_tmp_1761321727304.png  oua_classifier
Music       _sc_tmp_1761321713940.png  _sc_tmp_1761321720754.png  _sc_tmp_1761321727951.png
Notifications _sc_tmp_1761321714554.png  _sc_tmp_1761321721418.png  _sc_tmp_1761321728575.png
RES4B4LI:/ $
```

Step 6: Clean Up

1. Uninstall AFLogical OSE:

- Once data extraction is complete and verified, safely uninstall the app from the Android device:

adb uninstall com.viaforensics.android.aflogical

2. Disconnect the Device:

- Safely disconnect the Android device from your computer.

By following these steps, you successfully perform a logical extraction on an Android device using AFLogical OSE, a valuable technique in digital forensic investigations.

Result:

The experiment using Aflogical OSE was successfully performed. The tool efficiently extracted logical data such as contacts, messages, and media files from the mobile device, demonstrating its usefulness in mobile forensic data acquisition.